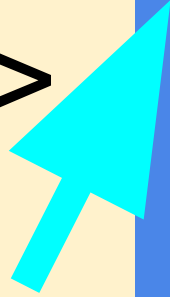


ONE PERSON IN GROUP-> CLICK USE TEMPLATE



WHEN THE COPIED TEMPLATE OF THESE SLIDES OPENS IN A NEW TAB:

- ADD YOUR Group NAME TO THE TITLE
- SHARE WITH GROUP MEMBERS (USE SHARE BUTTON)
- COPY THE TITLE AND LINK TO YOUR RECORDS (Digital folder/table of contents)
- Remember it is advisable to make a PERSONAL copy and link it when DONE

Solution Stoichiometry.

ESSENTIAL QUESTIONS

Remember, you should be able to thoroughly answer these AFTER learning through following resources.

- How can we calculate dilutions and concentrations we need in the lab?
- How can we safely prepare dilutions for our needs?

Instructions

Follow through the **Engage to Apply** resources in order .

- Once complete, choose from Extend resources as interested.
- Record your learning in your **these slides- Remember to keep track of where to find these slides for studying!**
- Complete the **Essential Questions Response in the Apply Section** .
- Try the Extend section and optional Apply practice

Hyperdoc Elements

Click for quick access to:

- [Engage](#)
- [Explore](#)
- [Apply](#)

Terminology - As you encounter terms new to you, define in your own words here, include examples (make them up!)



Part 1: ENGAGE

Explore - Part A: 5 minutes

Importance of Stoichiometry

Go to [Real Life Stoichiometry](#) and scroll down until you see "1. Airbags". Each group member choose a different application to read (air bags, rockets etc).

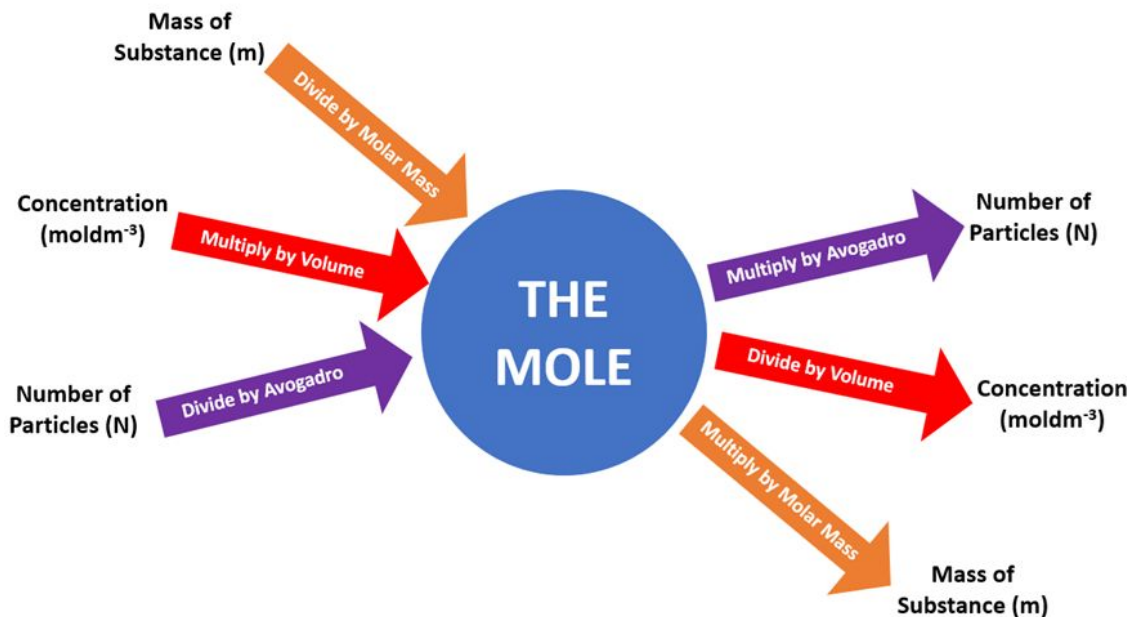
Share briefly with your group and explain what could happen if the stoichiometry is incorrectly applied in your application.

Part 2: EXPLORE

Explore - Solution Stoichiometry

Stoichiometric Calculations can be performed using **Solutions** with known volumes and concentrations

- These can be used to calculate the volumes, concentrations, moles or masses of other compounds

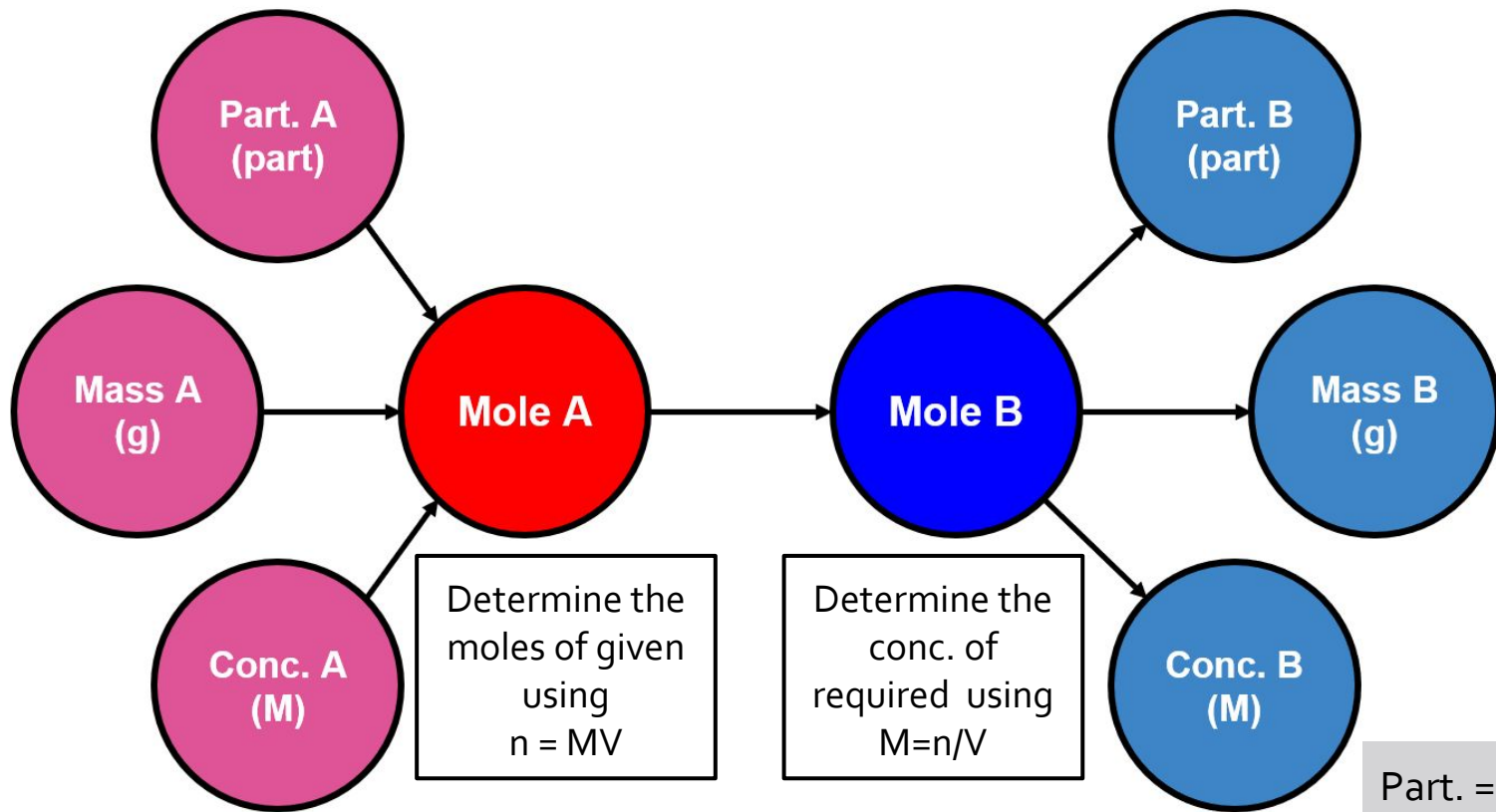


Explore - Solution Stoichiometry

Stoichiometry with solutions involves an extension to the problem solving map.

- You already know how to calculate the moles of given from the starting points of mass or number of particles.
- Now you can use the $M = n/V$ formula to calculate moles from a starting solution.
- See the next slide and discuss as a group to ensure you understand.
- Stoichiometry **remains the same**: You need to determine the *moles of the given*, use the *molar ratio* to determine the *moles of required* then convert to *what is asked for* (mass, particles or solution)

Explore - Solution Stoichiometry



Explore - Practice 1 - Concentration from Mass

*100.0mL of AgNO_3 (unknown concentration) is reacted with excess solid copper. After leaving the reaction for a few days, 1.65g of Silver was precipitated out. What was the original AgNO_3 concentration? (use a charge of +1 for the copper ion that forms) **Ans: 0.153M***

Try yourselves on a Whiteboard then remove this box to check your process.

* Hint start with a balanced chemical equation & list your givens!

Explore - Practice 2 - Limiting Reagent

Aluminium sulphate reacts with calcium nitrate. If 20.0mL of 0.0150M Aluminum Sulphate and 30.0mL of 0.0185M Calcium nitrate react, which is the limiting reagent? How much (mass) precipitate is formed?

***** Use your stoichiometry knowledge! Balance first :) If you are unable to get the correct answer, confer with another group and then your teacher.***

Move this box to check your answer after trying. Feel free to check steps with other groups as you go. Eg. Did you get the same product as the precipitate?

Explore - Practice 3 - "Fully React"

What volume of 0.125M HCl is required to fully react with 0.500g of CaCO_3 ?



Discuss what "fully react" means in terms of the molar ratios in a chemical equation. Ensure you are able to solve for the answer below!

Ans: 79.9mL

Explore - Practice 4 - Minimum Volume

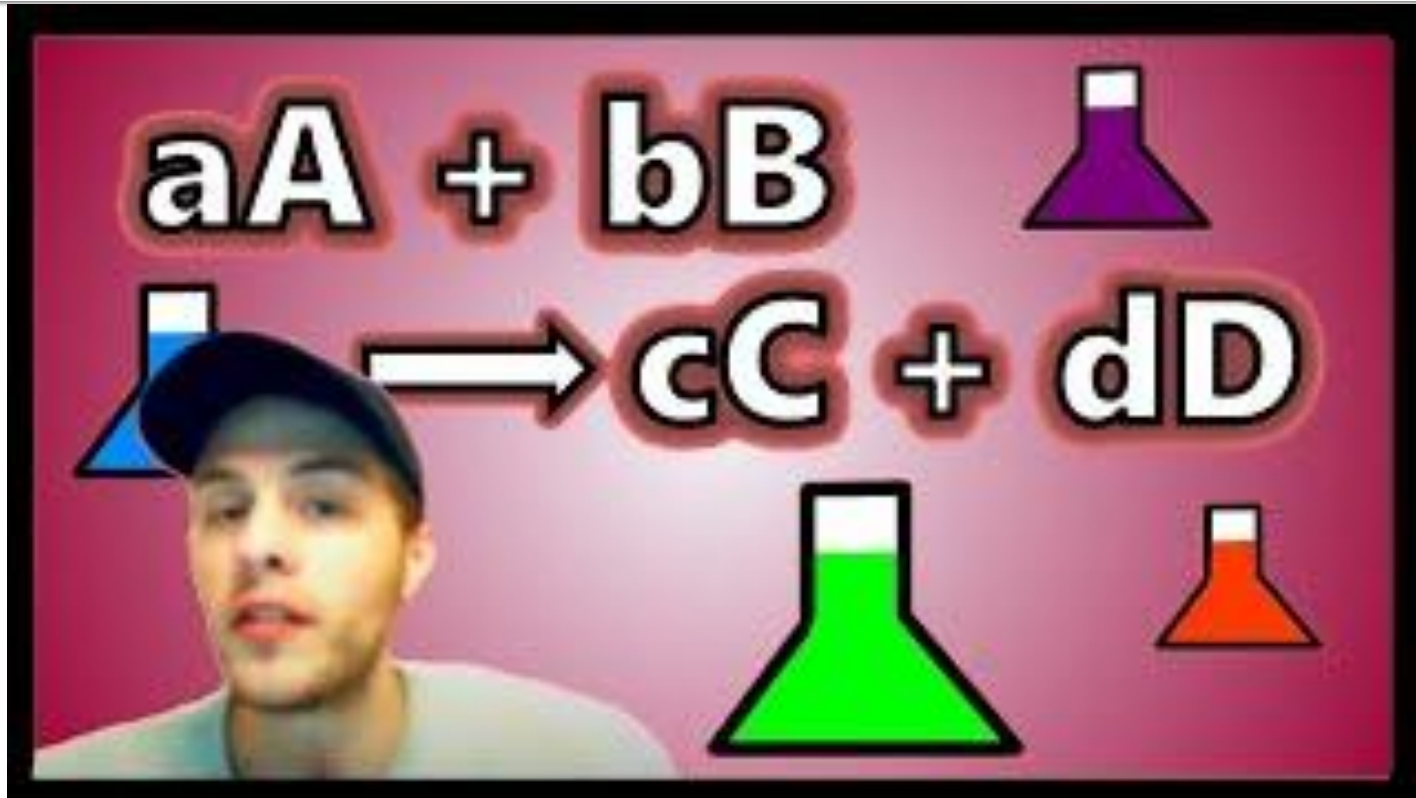
What volume of 0.4M HCl is needed to completely neutralize 47.1mL of 0.75M



Discuss the similarities and differences between this question and the previous (Practice 3) question. What steps are needed to solve this? Ensure you are able to solve for the answer below!

Ans: 176mL HCl

OPTIONAL Video help



Part 3: Apply

Apply- Complete the questions



Show all your work! Include formulas, conversions, units, etc.

Silver Chromate, Ag_2CrO_4 , is insoluble. Calculate the mass of silver chromate that forms when 50.0mL of 0.200 mol/L silver nitrate reacts with 50.0mL of 0.150 mol/L sodium chromate. **Ans: $m = 1.66\text{g}$**

Seek assistance from peers/teacher if you are unable to get the indicated answers to any problems.